

Cropping Plan and Rotation of a Family Farm

Introduction to Numerical Optimization - Project 2026

The problem

A farm of 80 hectares must decide, for each block of parcels, how many hectares to sow with each crop, how much water to pump and how many hours of labour to hire. The year is split into four periods.

The plan must satisfy the following.

1. **Land.** Each block may be divided between crops as finely as one likes, and the sum of the areas sown on a block must equal to its area. The crop `cover_fallow` is the non-productive option: it yields nothing and costs a little.
2. **Yields.** Each block has one soil class and the yield of a crop depends on that soil class.
3. **Water.** Irrigation is drawn from a farm reservoir, fed by an inflow in each period and by pumping from a stream. The reservoir starts at `reservoir_initial` and has a maximum level. The pump has a maximum volume per period, and the abstraction permit a maximum volume per year. Pumping costs money and the inflow is free. Any inflow in excess of the maximum level is lost.
4. **Reserve.** The reservoir level at the end of the last period must be at least `reservoir_terminal`.
5. **Labour.** Each period offers a fixed number of family hours. Extra hours are hired at that period's wage, up to that period's cap.
6. **Contracts.** Sugar beet must deliver at least `contract_min_t`. Sugar beet and potatoes are paid the contract price up to `contract_max_t` and the free-market price beyond it. Every other crop is sold at a single price, without limit.
7. **Diversification.** No crop may occupy more than its `max_farm_share` of the farm area in a year.
8. **Non-productive area.** At least `non_productive_share` of the farm area must be left non-productive.
9. **Rotation.** Three rules couple consecutive years; they are stated in Part C and do not apply in Parts A and B.

Objective. Maximize the gross margin: revenue, minus the variable cost of every hectare sown, minus the hired wage bill, minus the cost of pumping.

Data and conventions

File	Content
<code>blocks.csv</code>	Area and soil class of each block
<code>crops.csv</code>	Variable cost per hectare, diversification share, rotation share and crop type
<code>yields.csv</code>	Yield of each crop on each soil class
<code>calendar.csv</code>	Irrigation and labour requirement of each crop in each period
<code>labour.csv</code>	Family hours, hiring limit and wage in each period
<code>water.csv</code>	Reservoir inflow and pumping capacity in each period
<code>markets.csv</code>	Prices, contract minimum, contract ceiling and free-market price
<code>parameters.csv</code>	Reservoir, abstraction permit, diversification and rotation parameters
<code>initial_areas.csv</code>	Cropping plan of the year that just ended, used only in Part C

Table 1: Data files supplied with this statement

- Areas are in hectares, water in cubic metres, work in hours, output in tonnes, money in euros.

- The four periods `P1_spring`, `P2_early_summer`, `P3_late_summer` and `P4_autumn` partition the year.
- In `markets.csv`, a blank `contract_max_t` means the whole output of that crop is sold at `price_EUR_per_t`; a blank `contract_min_t` means there is nothing to deliver.

A The coming year

1. Formulate the one-year problem as a linear program. Define the sets, the parameters, the variables, the objective function and every family of constraints, and say in one or two sentences what each family expresses agronomically.
2. Implement and solve the model. Report the optimal gross margin and the cropping plan block by block. Show, with figures, how the plan uses the reservoir and the hired labour. Which resources are exhausted at the optimum, and which are not?

B What the plan is worth

1. Write your model in standard form $\min\{c^T x : Ax = b, x \geq 0\}$ and write the corresponding dual problem.
2. Interpret the dual variables of the land balances, of the reservoir balances and of the labour constraints, each in its own unit. Why is water worth nothing in one period and something in another?
3. Verify numerically that the primal and dual objective values agree.
4. Let L be the annual abstraction permit. Assume that L may be changed by a value ΔL . Give the interval of L outside of which the current optimal basis would change. Inside that interval the optimal plan is an affine function of L : say which areas move, in which direction and at what rate, and what happens to the gross margin once L leaves the interval on either side. Interpret the results.
5. Solve the model with the primal simplex and the dual simplex. Compare objective values, iteration counts and solution times, and comment on what the comparison is and is not evidence for at this size. Then add ten family hours in the autumn period and discuss which method is naturally suited to reoptimizing from the previous basis.

C The rotation

The horizon becomes three consecutive years, under the following conventions.

- The reservoir carries over from one year to the next. The abstraction permit applies to each year separately. The terminal reservoir requirement applies at the end of the third year only.
- Gross margins are discounted at the rate r given in `parameters.csv`: the gross margin of year $y \in \{1, 2, 3\}$ is weighted by $1/(1+r)^{y-1}$, so year 1 is undiscounted and the objective is the sum of these weighted yearly margins.
- `initial_areas.csv` holds the plan of the year that has just ended and counts as year 0 in the rules below. It is history: it enters those rules as data and is not required to satisfy them.
- The three years face the same conditions: yields, prices, costs, inflows, family hours and hiring limits are those of the data files in each year.

Three agronomic rules are added.

- **Return period.** For each block b and each pair of consecutive years, the areas a crop occupies in those two years must add up to at most its `return_share` of the area A_b of that block — not of the two years' areas taken together.
- **Preceding crop.** On each block, the potato area of a year cannot exceed the cereal area of that block in the previous year.
- **Legume.** Legumes must cover at least `legume_share` of the farm area in every year.

1. Extend your model to three years and add the three rules. Write them down explicitly, taking care of how year 0 enters. Solve, and present the three cropping plans together with the discounted total gross margin.
2. Solve the same three years *without* the three agronomic rules and compare. What does the rotation cost the farm over three years, in euros and as a percentage? Then remove the rules one at a time to find which one is the expensive one, and explain agronomically why the plan resists it.
3. The farm has money to invest. Propose three investments it could make, one of them the rental of an extra parcel of land. For each investment, say which parameter it changes and by how much, re-solve the three years, and report the gain in the discounted gross margin. Any cost or lifetime you need is not in the data files: state your assumptions explicitly. Conclude, for each investment, whether it is worth making.

Deliverables

Work by groups of two students or individually. Send an email to Laurie Boveroux (laurie.boveroux@uliege.be) with the names of the group members by October 7, 2026.

Submit on Gradescope your report and code by November 2, 2026 at 23:59. The report must be in PDF format and cannot exceed 8 pages in total. Your code must be written in Julia and it must read the nine CSV files of Table 1 from a `data` subdirectory.

Be ready to present your results on November 4, 2026. **No second session will be organised for the project**, the grade is final.